José Carlos Ruiz Sánchez

Full-stack Developer

Full-stack developer who has taken a real-time geospatial data platform to production with Node.js, Express, TypeScript, Next.js, PostgreSQL/PostGIS and Docker, covering everything from data ingestion to the frontend, testing and deployment.

Málaga, Spain | +34 608 25 75 74 | josecarlosruizsan@gmail.com
linkedin.com/in/jose-carlos-ruiz-sanchez | github.com/jrs407

> Portfolio with details on every role and project: **jrs407.github.io/web-personal**

TECHNICAL SKILLS

Programming languages	JavaScript, TypeScript, Python, SQL
Backend	Node.js, Express.js, REST APIs, OpenAPI/Swagger, Python/Flask, microservices, API Gateway
Frontend	React, Next.js, Vite, React Router, Leaflet, D3.js, Recharts, Tailwind CSS, CSS Modules, HTML5 Canvas API
Databases and data	PostgreSQL, PostGIS (spatial queries), MySQL, n8n (scheduled ingestion, webhooks, API integration)
Testing	Vitest, Jest, Supertest, React Testing Library, unit and integration testing
DevOps	Docker, Docker Compose, multi-stage builds, Nginx (reverse proxy, API Gateway), Git (feature branches, pull requests, code review)
Security	JWT, bcrypt, helmet, rate limiting, CORS, HMAC pseudonymization
Cloud	OpenStack, Google Cloud (Compute Engine, Cloud Run, Cloud SQL, App Engine, Artifact Registry), Azure, Vercel; Ubuntu virtual machines, SSH, networking and firewall rules (Master's degree labs)
Academic knowledge	Java, Spring, JUnit, Angular, MongoDB, Apache Spark, Cassandra, Power BI, Scrum, Kanban, ISO 27001, ISO 9001, GDPR

PROFESSIONAL EXPERIENCE

Full-stack Developer (Internship)

Mar 2026 – Sep 2026

Applied Computing Group (ACG) · University of Almería · Extracurricular internship, 25 h/week, on-site and remote

- Developed UAL Trace, a web application that maps estimated campus occupancy from devices connected to the Wi-Fi network (Cisco Meraki), with data refreshed every 2 minutes. Led most of the development, taking it from a prototype with mock data to production.
- Built a REST API with Node.js, Express.js and TypeScript: around 30 operations documented with OpenAPI/Swagger and covered by more than 200 tests with Vitest and Supertest.
- Designed the PostgreSQL/PostGIS data model (12 tables, spatial and time-based indexes, migrations) and an n8n ingestion pipeline processing about 3,000 rows every 2 minutes, with HMAC-SHA512 pseudonymization and nearest-neighbor assignment to a hexagonal grid.
- Developed the Next.js and React frontend (about 80% of the code is mine) with Leaflet and D3.js: density, building occupancy and inter-building route layers, a historical mode with playback, PDF reports, mobile support and tests with React Testing Library.
- Optimized performance under real usage: in-memory cache that merges concurrent identical requests, multi-process backend with the Node.js cluster module, covering indexes, and an endpoint that replaced 13 sequential requests with a single one.
- Deployed with Docker (multi-stage builds, non-root user) behind an Nginx reverse proxy. Coordinated daily with the group lead through feature branches and code reviews.

PROJECTS

Tokimori · Full-stack microservices web application

Feb 2026 – Aug 2026

Personal project · Private repository (access on request) · Demo: youtube.com/watch?v=hSbAzT15mxI

- 6 microservices in Node.js, Express 5 and TypeScript over MySQL, behind an Nginx API Gateway acting as the single public entry point.
- JWT authentication with secure cookies, bcrypt, helmet, rate limiting and CORS. Tests with Vitest and Testing Library on the frontend and Jest and Supertest on the backend.
- Frontend in React 19, Vite 7, React Router 7 and CSS Modules, with a visual editor built on the HTML5 Canvas API without libraries (layers, rotation, undo/redo), drag and drop with FLIP animations and custom i18n.
- Docker Compose with a production configuration and a development override with hot reload.

Auscultify · Bachelor's Thesis (9.4/10)

Dec 2024 – Jan 2026

Web application for medical auscultation training with real audio recordings · github.com/jrs407/Auscultify

- 9 microservices (8 in Node.js, Express and TypeScript and 1 in Python and Flask) with MySQL 8 and Docker Compose. Passwords hashed with bcrypt.
- Recommender system with 11 algorithms, from baseline methods to user-based and item-based collaborative filtering using Pearson correlation.
- Frontend in React 19, TypeScript and Vite with Chart.js statistics, social features and an admin panel. Analysis with 27 functional requirements, Figma prototypes and documentation written in LaTeX.

PuppyClicker · Full-stack browser game

Apr 2025 – Jun 2025

Personal project · github.com/jrs407/PuppyClicker

- REST API with 10 endpoints in Node.js, Express 5 and TypeScript over MySQL 8 (7 tables), with JWT, bcrypt and all game logic computed server-side.
- Purchases run in transactions with SELECT … FOR UPDATE. React 19 and Vite frontend, Docker Compose deployment and feature-branch workflow merged through 8 pull requests.

Personal portfolio website Sep 2026 – Present

Personal project · github.com/jrs407/web-personal

- 11-page static site in HTML, CSS and JavaScript with no frameworks, bilingual with custom i18n, light/dark theme, accessibility and SEO, deployed on GitHub Pages.

EDUCATION

Master's Degree in Computer Engineering 2025 – Present

University of Almería · Big Data specialization · 100% online, compatible with full-time work

- Cloud labs: virtual machines on OpenStack, Google Cloud and Azure; applications on Vercel and Google App Engine with Cloud SQL; containers on Compute Engine and Cloud Run using Docker Hub and Artifact Registry.

Bachelor's Degree in Computer Engineering 2021 – 2026

University of Almería · Double specialization in Software Engineering and Information Systems

- Bachelor's thesis (Auscultify) graded 9.4/10, defended in February 2026.
- Consulting project with a real company: risk analysis, contingency plans, ISO 27001 and GDPR compliance. Team projects using Scrum and Kanban.

LANGUAGES AND AVAILABILITY

Languages: Spanish (native) | English B2 (Cambridge English: First, FCE)

Availability: Immediate start | Full-time | On-site, hybrid or remote | Willing to relocate anywhere in Spain